Disentangling Pragmatic and Structural Cues in L2 Relative Clause Processing: An Eye-Tracking Study of Mandarin-English Bilinguals

In English, subject-modifying RCs create a center-embedded (CE) structure while object-modifying RCs form a right-branching (RB) structure. Previous research has focused on the processing asymmetry between subject-extracted RCs (SRCs) and object-extracted RCs (ORCs), but much less research has focused on how differently CE RCs and RB RCs are processed. Since most sentence processing hypotheses were established based on the RC type asymmetry, these hypotheses yield distinct predictions regarding RC modification. The syntactic-based perspective shift hypothesis [1] proposes that structures maintaining the subject's perspective are easier to process than those shifting it. Memory-based hypotheses, including dependency locality theory (DLT) [2] and structural distance hypothesis [3], predict processing difficulty for embeddedness. Moreover, the discourse function of restrictive RCs may also play a role in the L1 processing. This pragmatic effect was found in the residual reading time at the RCs [2], with the RB RCs requiring more time than CE ones [2]. However, when it comes to L2 processing, the role of pragmatics is less understood. The goal of this study is to investigate L2 readers' sensitivity to both pragmatic cues and structural complexity. L2 processing is hypothesized to be less sensitive than L1's to the functional application of restrictive RCs within the discourse structure, especially if the L2 is learned in an environment where the language is not dominant. Furthermore, the structural complexity may impose greater processing load for L2 readers than L1 readers. With eye movement indices, we also aim to disentangle the influence of different cues to further unveil the dynamics of sentence processing.

The present study adopted a 2 x 2 design manipulating RC modification (CE vs. RB) and RC types (SRC vs. ORC) in an eye-tracking sentence reading experiment (Table 1). To control the semantic effect of thematic-role relationship, all the NPs are humans and, in terms of plausibility, they form reversible relationships with the VPs both in the RCs and in the matrix clauses. Four lists, each consisting of 40 target items and 96 filler sentences, were created using a Latin square design. Forty native Mandarin Chinese speakers learning English as their L2 participated in this experiment. As Mandarin Chinese employs prenominal RCs, which differ structurally from English's postnominal RCs, direct transfer of L1-based processing procedures is not expected [3] for these participants.

Linear mixed-effects models were fitted using for the analysis of eye movement indices at the RC region, and at the NPs and VPs in different clauses. The results of total viewing time on the RC region showed that CE structures required greater processing effort than RB ones and ORCs took longer time than SRCs, but no interaction was found (Figure 1). Similar patterns were also found in the number of fixations per sentence, the total time spent per sentence, and the rereading time at the NPs of RC. These results consistently suggested a greater cause of storage and integration load based on structural complexity and supported the memory-based hypotheses. The interaction was only found on the first-pass time at the NP in the RC (Figure 2), suggesting that the CE-SRCs are easier to process than the other conditions during an earlier stage. This finding conforms to the syntactic-based perspective shift hypothesis [4]: When integrating the NP in the RC, only the CE-SRCs are without a perspective shift. However, at the same region, the total viewing time showed only a SRC preference, and the rereading time showed two main effects without interaction. It seems that though the perspective shift influences L2 processing on this critical NP of the clause during the early-stage recognition and integration, the later-stage reanalysis is mainly a function of structural complexity. In sum, L2 learners are less sensitive to discourse functions, and their syntactic integration and memory resources may be greatly limited by structural complexity.

Table 1. Example sentence set and regions of interest for analyses of the effects of RC Modification (CE: center-embedded vs. RB: right-branching) and RC Type (SRC vs. ORC)

Conditions	Example Stimuli (Region)
CE-SRC	The student [**who** $_{gap_s}$ **directly challenged the professor**] trusted the principal.
	(Head-N) [RC:(REL) (ADV) (VP) (NP)] (Mx-VP) (Mx-NP)
CE-ORC	The student [**who the professor directly challenged** $_{gap_o}$] trusted the principal.
	(Head-N) [RC: (REL) (NP) (ADV) (VP)] (Mx-VP) (Mx-NP)
RB-SRC	The principal trusted the student [**who** $_{gap_s}$ **directly challenged the professor**].
	(Mx-NP) (Mx-NP) (Head-N) [RC: (REL) (ADV) (VP) (NP)]
RB-ORC	The principal trusted the student [**who the professor directly challenged** $_{gap_o}$].
	(Mx-NP) (Mx-NP) (Head-N) [RC: (REL) (NP) (ADV) (VP)]

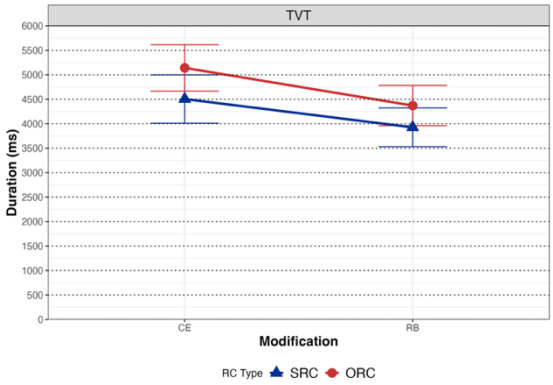

Figure 1. Total viewing time (TVT) on the RC region as a function of RC modification and RC type.

Both the modification effect (t = -4.538, p <.001) and the RC type effect (t = 4.527, p <.001) were significant. No interaction was found (t = -0.928). Error bars represent 95% confidence intervals.

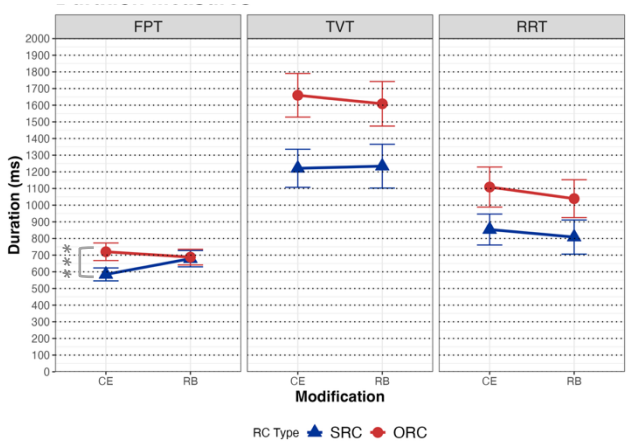

Figure 2. First-pass time (FPT), total viewing time (TVT), and rereading time (RRT) on the NP in the RC as a function of RC modification and RC type.

FPT: While there was no modification effect (t = 0.681, p =.353), RC type effect was significant (t = 3.166, p <.01). The interaction was significant (t = -2.156, p <.05), with the SRC significantly shorter than ORC (t = 3.75, p <.01) in the CE modification.

TVT: Only RC type effect was found (t = 7.824, p <.001). There was no modification effect (t = -1.157, p =.247) or interaction (t = 0.339, p =.734).

RRT: Two main effects were significant (modification: t = -2.432, p <.05; RC type: t = 5.361, p <.001). There was no interaction (t = 0.66, p =.509).

Error bars represent 95% confidence intervals.

References:

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